polarized lenses: when the photon hits the glass, it randomly "chooses" between reflecting back and passing through. The astounding thing is that such a photon can have a partner photon that has sped miles away in the opposite direction and yet, when confronted with the same 50–50 probability of passing through another polarized sunglass lens, will somehow do whatever the initial photon does. *Even though each outcome is determined randomly and even though the photons are far apart in space, if one photon passes through, so will the other.* This is the kind of nonlocality predicted by quantum mechanics.

Einstein, who was never a great fan of quantum mechanics, was loath to accept that the universe operated according to such bizarre rules. He championed more conventional explanations that did away with the notion that particles randomly select attributes and outcomes when measured. Instead, Einstein argued that if two widely separated particles are observed to share certain attributes, this is not evidence of some mysterious quantum connection instantaneously correlating their properties. Rather, just as Scully argued that the spheres do not randomly choose between red and blue, but instead are programmed to flash one particular color when observed, Einstein claimed that particles do not randomly choose between having one feature or another but, instead, are similarly "programmed" to have one particular, definite feature when suitably measured. The correlation between the behavior of widely separated photons is evidence, Einstein claimed, that the photons were endowed with identical properties when emitted, not that they are subject to some bizarre long-distance quantum entanglement.

For close to five decades, the issue of who was right—Einstein or the supporters of quantum mechanics—was left unresolved because, as we shall see, the debate became much like that between Scully and Mulder: any attempt to disprove the proposed strange quantum mechanical connections and leave intact Einstein's more conventional view ran afoul of the claim that the experiments themselves would necessarily contaminate the very features they were trying to study. All this changed in the 1960s. With a stunning insight, the Irish physicist John Bell showed that the issue could be settled experimentally, and by the 1980s it was. The most straightforward reading of the data is that Einstein was wrong and there can be strange, weird, and "spooky" quantum connections between things over here and things over there.[5]